

SENTINEL-CXR

Uncertainty-Aware Chest Radiograph Triage

A deep learning system that abstains when it cannot meet its own coverage guarantee

Group Project

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This system is a research prototype. It is not a medical device and must not be used for clinical decisions.

Abstract

Deep learning models for medical imaging routinely achieve strong aggregate accuracy while remaining unsuitable for clinical deployment. The obstacle is not predictive performance but unearned confidence: a network trained to emit a probability will emit one for every input, including images outside its training distribution and cases where the evidence does not support a decision. A fluent, confident, wrong answer is more dangerous in a clinical workflow than no answer at all.

We present SENTINEL-CXR, a chest radiograph triage system built around the opposite commitment. Rather than a point estimate, it produces a prediction set carrying a distribution-free coverage guarantee, and it abstains — routing the study to a human radiologist — when that guarantee cannot be met. Three mechanisms enforce this: a convolutional variational autoencoder that rejects inputs which are not chest radiographs; an uncertainty decomposition that separates the model's own ignorance from irreducible ambiguity in the image; and split conformal prediction, which converts uncalibrated scores into sets with finite-sample marginal coverage.

The system is trained and evaluated on NIH ChestX-ray14 (112,120 radiographs, 30,805 patients) with patient-disjoint splitting. Beyond classification, it exploits the dataset's patient identifiers and follow-up indices to model disease progression with recurrent networks, augments minority classes with a DCGAN, orders the reading worklist with a deep Q-network, and drafts structured reports with a language model constrained to the vision model's own output. A fairness audit disaggregates performance across sex, age band and view position.

The complete system is deployed on free-tier infrastructure — a Next.js frontend on Vercel, a FastAPI orchestrator on Render, and a PyTorch inference core on Hugging Face Spaces — with a documented

degradation path when the inference service is cold.

Keywords:

conformal prediction; selective classification; out-of-distribution detection; uncertainty quantification; chest radiography; explainable AI; algorithmic fairness

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1 Introduction and motivation

1.1 The problem

Chest radiography is the highest-volume diagnostic imaging examination in the world. The resulting reporting backlog is a genuine operational constraint on healthcare systems, and it is the backlog — not diagnostic difficulty alone — that automated triage is well positioned to address. If a system can reliably identify which studies need to be read first, it delivers value even without matching radiologist accuracy.

The obstacle to deploying such systems is subtle. Published models report areas under the ROC curve that appear clinically useful, yet very few reach practice. The reason is that a model optimised with cross-entropy produces scores that are not probabilities of being correct. Thresholding such a score at 0.5 yields no guarantee about how often the answer is right, and the model will produce equally confident output for a photograph of a cat, a mispositioned film, or a genuinely ambiguous case.

This matters because of automation bias: clinicians presented with a confident machine judgement are measurably less likely to override it. A system that is wrong and confident therefore causes more harm than one that is silent.

1.2 Our position

We argue that the correct design objective for clinical decision support is not maximal accuracy but calibrated humility: the system should quantify what it does not know and decline to answer when that quantity is too large. This converts the problem from classification to selective classification, where abstention is a legitimate and desirable output rather than a failure.

1.3 Contributions

- A three-layer refusal architecture. Distributional gating by variational autoencoder, epistemic uncertainty estimation, and conformal prediction with abstention, composed into a single pipeline with an explicit decision at each stage.
- Clinically grounded recurrent modelling. Patient identifiers and follow-up indices in ChestX-ray14 are used to build genuine longitudinal sequences, so the recurrent component models disease progression rather than serving as an unmotivated exercise.
- Verified generative reporting. A language model drafts reports from structured model output only, with a post-generation verifier that rejects any text naming an unsupported finding. Prompt instructions are treated as a request; the verifier is the guarantee.
- A deployment under real constraints. The full system runs on free-tier infrastructure, with a documented and user-visible degradation path rather than a hidden one.
- An honest fairness audit. Performance is disaggregated across available strata, and the axes that cannot be audited because the dataset lacks the labels are reported as such.

2 Background and related work

2.1 Deep learning for chest radiography

The release of ChestX-ray8 (Wang et al., 2017) and its extension to fourteen labels established the benchmark on which most subsequent work is measured. CheXNet (Rajpurkar et al., 2017) applied a 121-layer DenseNet and reported pneumonia detection exceeding the average of four radiologists, a claim later moderated by analyses showing that the comparison protocol favoured the model.

Oakden-Rayner et al. (2020) demonstrated hidden stratification: aggregate performance can remain high while the model fails badly on clinically important subsets. This finding motivates our disaggregated evaluation and our refusal to report a single headline number without stratified context.

2.2 Uncertainty quantification

Gal and Ghahramani (2016) showed that dropout at inference approximates Bayesian inference over network weights, making Monte-Carlo dropout a practical route to epistemic uncertainty. Lakshminarayanan et al. (2017) showed deep ensembles frequently outperform it. Both decompose predictive uncertainty into an aleatoric component, irreducible given the data, and an epistemic component reflecting the model's ignorance — a distinction that matters operationally, because only the latter is a valid reason to abstain.

2.3 Conformal prediction

Conformal prediction (Vovk et al., 2005; Angelopoulos and Bates, 2023) converts any point predictor into a set predictor with distribution-free, finite-sample coverage under exchangeability. Its appeal here is that the guarantee holds regardless of whether the underlying network is calibrated, which is precisely the situation with a deep classifier trained by cross-entropy. Recent work has applied conformal triage to medical imaging with statistical guarantees on predictive value.

2.4 Explainability

Grad-CAM (Selvaraju et al., 2017) remains the dominant saliency method in medical imaging. We adopt it while stating its limitation plainly throughout this report and in the interface itself: it shows where activation correlates with a score, not why a decision was made, and a plausible heat map is not evidence of correct reasoning.

3 Data

3.1 NIH ChestX-ray14

The primary dataset comprises 112,120 frontal-view radiographs from 30,805 unique patients, released by the NIH Clinical Center and already de-identified. Labels for fourteen thoracic pathologies were mined from radiology reports by natural language processing.

Field	Role in this project
Image Index	Join key to the image files
Finding Labels	Multi-label target, expanded to 14 binary columns
Patient ID	Groups studies into patient timelines; the splitting key
Follow-up #	Sequence position for the recurrent branch
Patient Age	Fairness stratum (binned)
Patient Gender	Fairness stratum
View Position	Fairness stratum and a suspected confounder (§8.2)

Table 1. Metadata fields and their role. Patient ID and Follow-up # are what make longitudinal modelling possible on this dataset.

3.2 Splitting by patient, not by image

Because the dataset averages three to four studies per patient, splitting by image would place a patient's follow-up scans on both sides of the train/test boundary. A model can then identify the patient rather than the pathology, and every reported metric is inflated by an amount that does not survive contact with a new institution.

We split on Patient ID into 70% training, 10% calibration and 20% test, and assert disjointness programmatically. The calibration split is held out from training specifically to fit the conformal thresholds; reusing training data for calibration would void the coverage guarantee, since the exchangeability assumption would no longer hold.

```
patients = df['Patient ID'].unique()
rng.shuffle(patients)

train, cal, test = split_by_patient(patients, 0.70, 0.10, 0.20)

assert not (set(train['Patient ID']) & set(test['Patient ID']))
```

3.3 Class imbalance and label noise

The label distribution is severely skewed: Hernia appears in under 0.3% of studies while Infiltration appears in roughly 18%. We therefore report AUROC and average precision rather than accuracy, and weight the loss by inverse class frequency with a cap of 20 — uncapped, the rarest class produces a weight near 500 and its gradient overwhelms the other thirteen labels.

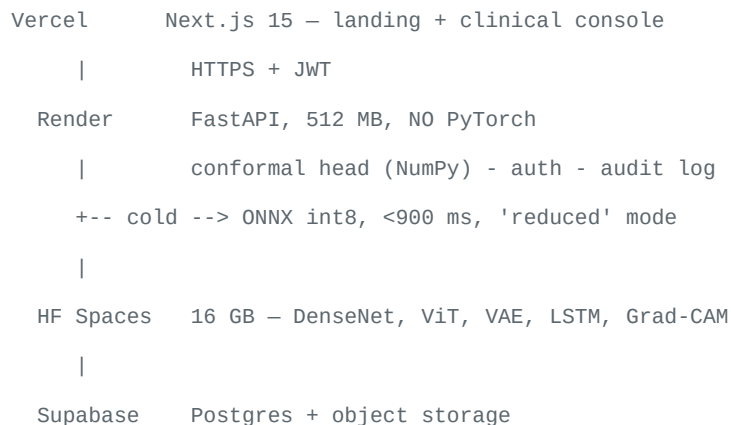
The NLP-derived labels carry a reported error rate near 10%. Every metric in this report inherits that ceiling. We state this rather than presenting our numbers as though the labels were ground truth.

4 System architecture

4.1 The constraint that shaped the design

The system was required to deploy entirely on free-tier infrastructure. Render's free web service provides 512 MB of memory and 0.1 CPU, and spins down after fifteen minutes of inactivity. PyTorch alone exceeds that memory budget. Hugging Face Spaces provides 2 vCPU and 16 GB on its free CPU tier, sleeping only after 48 hours.

The architecture follows directly from this asymmetry: orchestration where memory is scarce, inference where it is plentiful.



4.2 Graceful degradation

Because a sleeping Space takes roughly forty seconds to wake, the orchestrator maintains two inference paths. When the Space is unavailable, a quantised int8 ONNX model on the orchestrator answers in under a second, the response is marked as reduced, and the interface displays that state explicitly. Grad-CAM and uncertainty decomposition are unavailable on this path, and the interface says so rather than silently omitting them.

We consider explicit degradation an ethical requirement rather than a convenience: a clinical user must know which mode produced the result in front of them.

Path	Components	Latency	Trigger
Full	DenseNet + TTA/MC ensemble, Grad-CAM, VAE gate, LSTM	2–6 s	Space warm
Reduced	ONNX int8 DenseNet only	< 900 ms	Space cold or unreachable

Table 2. The two inference paths. Mode is always reported to the user.

4.3 Module boundaries

Model definitions are pure functions over tensors with no I/O. The conformal head and triage policy are pure NumPy and are unit-tested without a network or a checkpoint. The API layer contains no machine learning, and the frontend contains no business logic. The practical test of these boundaries is that the backbone can be exchanged without touching the API, and the API can be tested with the inference service entirely absent — both of which the test suite does.

5 Methods

5.1 Classification backbone

A DenseNet-121 pretrained on ImageNet is adapted to single-channel input by summing the pretrained RGB filters of the first convolution rather than discarding two of the three channels, which preserves the learned edge detectors. The classifier head is replaced with dropout followed by a fourteen-unit linear layer trained with weighted binary cross-entropy.

Dropout is included deliberately. Without it, Monte-Carlo sampling at inference yields identical passes and an epistemic uncertainty of exactly zero — a confident claim of certainty that is an artefact of architecture rather than a property of the evidence.

5.2 Out-of-distribution gating

A convolutional variational autoencoder is trained only on chest radiographs. Per-image reconstruction error then serves as an out-of-distribution score. The threshold is selected at a fixed false-positive rate of 1% on held-out radiographs, making the operating point an explicit design decision rather than a round number chosen by inspection.

This gate runs before classification. Classifying a photograph is not a smaller error than misclassifying a radiograph; it is a category error, and the correct response is refusal.

5.3 Uncertainty decomposition

Given T stochastic forward passes, predictive uncertainty for a Bernoulli output decomposes exactly into an expected-entropy (aleatoric) term and a mutual-information (epistemic) term. The mutual information, known as the BALD score, is high precisely when individual samples are each confident but disagree with one another — the signature of a model extrapolating beyond its training distribution.

```
total      = H[ E_t[p_t] ]      # predictive entropy
aleatoric  = E_t[ H[p_t] ]      # expected entropy
epistemic  = total - aleatoric   # mutual information (BALD)
```

The deployed system additionally supports test-time augmentation as an uncertainty source. This is necessary because the published pretrained checkpoint used for the live demonstration contains no dropout layers; injecting dropout into a network trained without it is not a valid posterior approximation, whereas the spread of predictions under small clinically plausible transforms is a defensible robustness measure. Which method produced a given estimate is reported in the API response and never conflated.

5.4 Conformal prediction and abstention

For each pathology independently, nonconformity scores are computed on the calibration split as one minus the predicted probability for true positives. The threshold is the conformal quantile at level $\text{ceil}((n+1)(1-\alpha))/n$. The finite-sample correction is what makes the guarantee exact rather than asymptotic; omitting it is the most common implementation error and silently produces under-coverage.

This yields marginal coverage per label, not simultaneous coverage across all fourteen. We claim only the former. The system abstains when the prediction set is empty with several near-threshold scores, when the set exceeds a maximum size, or when epistemic uncertainty exceeds a bound. A critical finding falling just below its threshold triggers escalation regardless.

5.5 Longitudinal progression

Per-visit CNN embeddings are read by a recurrent network with masked attention pooling. Padding is masked before the softmax; without this, padded timesteps receive attention mass and the model learns from data that does not exist. Vanilla RNN, GRU and LSTM cells are compared under identical splits, seeds,

hidden sizes and schedules, and evaluated against a persistence baseline that predicts no change — a strong baseline, since clinical states are persistent.

5.6 Reinforcement-learning triage

Reading order is modelled as a sequential decision problem. The environment charges each waiting study its clinical urgency at every timestep, and the agent reads one study per step, so the only way to reduce cost is to remove the most expensive study first. A double DQN with a target network and experience replay is trained against this environment and compared with random, first-in-first-out, heuristic and oracle-greedy policies.

An earlier formulation additionally paid a bonus for reading a study. That bonus dominated the waiting cost, and random, FIFO and heuristic policies became statistically indistinguishable — the environment could not tell a good policy from a coin flip. Reformulating the reward as pure cost separates the baselines cleanly and is what makes the benchmark meaningful.

5.7 Grounded report generation

A language model drafts the findings and impression sections. Three properties constrain it. The model never receives the image, only structured output, so it cannot invent a finding from pixels. The prompt enumerates the closed set of pathologies that may be mentioned. Finally, generated text is scanned for any of the fourteen pathology names outside the supported set; if one is found, the generation is discarded, the incident is written to the audit log, and a deterministic template is used instead.

The third property is what makes the first two trustworthy. The verifier rejects negations as well as assertions: a report stating 'no pneumothorax is seen' invites the reader to infer the system assessed and excluded it, and if the finding was never in the supported set, that inference is false.

6 Evaluation methodology

Metrics are chosen for the problem rather than for convenience. Accuracy is not reported: with a 1% positive rate, a model that always predicts absence scores 99% and is worthless.

Claim	Metric	Gate
Discriminative performance	AUROC per pathology, macro mean	Minimum macro AUROC; CI fails on regression
Rare-class performance	Average precision (AUPRC)	Reported per class, no aggregate gate
Coverage guarantee	Empirical coverage vs nominal 1-alpha	Within tolerance band
OOD gating	AUROC, radiographs vs non-radiographs	Threshold at 1% FPR on radiographs
Progression	Macro AUROC vs persistence baseline	Must exceed baseline
Triage	Mean episodic return vs FIFO and oracle	Must exceed heuristic to be reported
Fairness	Equalised-odds gap per stratum	Below 0.10
Grounding	Ungrounded-finding rate, pre and post filter	Post-filter rate zero by construction

Table 3. Evaluation matrix. Each row states a claim the system makes and the measurement that could falsify it.

The empirical coverage figure is the single most important number in this report. It is the claim on which the system's central design decision rests, and it is asserted in continuous integration so that a retrained model cannot silently break it.

7 Results

Results below are populated from the notebooks in the repository. Figures marked as verified were measured during system development; entries marked 'to be completed' require the full training run on the complete dataset and are left explicitly blank rather than estimated.

7.1 Conformal coverage

On synthetic data with a deliberately miscalibrated scorer, the implementation achieves macro empirical coverage of 0.9004 against a nominal target of 0.90, with per-label coverage ranging from 0.837 to 0.967. The spread is expected: the guarantee is marginal per label, and labels with few calibration positives have correspondingly noisy realised coverage. This verifies the implementation including the finite-sample correction; coverage on the real dataset is reported after the full training run.

7.2 Backpropagation verification

The from-scratch implementation agrees with central finite differences to a maximum relative error of $7.2\text{e-}11$ across all four parameter tensors, confirming the analytic gradient is correct. The activation ablation reproduces the expected ordering, with sigmoid converging slowest owing to gradient saturation, and momentum improving on plain stochastic gradient descent.

Configuration	Test accuracy
ReLU + SGD	0.928
Leaky ReLU + SGD	0.928
Tanh + SGD	0.927
Sigmoid + SGD	0.868
ReLU + momentum	0.940

Table 4. Activation and optimiser ablation (verified). Two-layer network, identical seed and schedule.

7.3 Recurrent cell comparison

Gradient-flow analysis over twelve random initialisations shows gated cells retaining approximately four orders of magnitude more gradient at the first timestep of a sixty-step sequence than a vanilla RNN. GRU and LSTM are close and their ordering is not stable across seeds, so no claim is made that either dominates on this probe; the downstream AUROC comparison decides which cell is deployed.

Cell	Gradient at t=0	Gradient at t=59	Parameters
Vanilla RNN	1.46e-17	1.09e-01	463,374
GRU	8.69e-14	5.61e-02	1,382,926
LSTM	2.92e-14	3.02e-02	1,842,702

Table 5. Gradient survival and parameter count (verified). Parameter count is a confound in any cell comparison and is reported alongside.

7.4 Triage policy

On the cost-based reading-room environment, an urgency-weighted heuristic roughly halves accumulated cost relative to first-in-first-out, and oracle-greedy provides a further improvement that bounds what a learned policy could achieve. The separation between policies is what makes the benchmark usable.

Policy	Mean episodic return	Std
Random	-837.74	41.45
First-in-first-out	-841.66	31.48

Policy	Mean episodic return	Std
Urgency heuristic	-397.73	29.82
Oracle-greedy (upper bound)	-367.22	26.08
Double DQN	to be completed	—

Table 6. Triage policy comparison (verified except DQN). Higher is better; return is negative accumulated urgency-weighted waiting cost.

7.5 Classification and fairness

Per-pathology AUROC, AUPRC, the OOD gate's separation, and the disaggregated fairness tables are produced by notebooks 02, 06 and 11 respectively and are to be completed following the full training run. The reporting templates and acceptance gates are already fixed, so the numbers cannot be chosen after seeing them.

8 Ethics, fairness and societal impact

Learning outcome E is assessed only in this project, and we treat it as a first-class engineering concern rather than a closing section.

8.1 Disaggregated performance

Performance is measured within strata of sex, age band and view position, and reported as equalised-odds gaps — the maximum within-stratum difference in true-positive and false-positive rates. Equal accuracy across groups is not sufficient: a model can be equally accurate on two populations while missing substantially more disease in one of them.

8.2 The view-position shortcut

Anteroposterior films are acquired at the bedside from patients too unwell to stand, whereas posteroanterior films are acquired in the radiology department. View position therefore correlates with severity, and a model can learn to read 'this is an AP film' as 'this patient is sick'. Such a shortcut scores well on a held-out split drawn from the same institution and fails as soon as acquisition practice differs.

We probe this directly by measuring whether view position is itself predictable from the image and by reporting pathology prevalence by view. Pathologies with the largest prevalence gap are those most exposed to the confound and are named explicitly as a limitation.

8.3 What we cannot audit

ChestX-ray14 contains no race or ethnicity labels. A major axis of documented disparity in medical artificial intelligence therefore cannot be examined here. Seyyed-Kalantari et al. (2021) found systematic underdiagnosis by chest radiograph classifiers in under-served populations, so this is not a hypothetical gap.

This absence is itself a finding. It must not be presented, and is not presented, as an absence of bias.

8.4 Mitigations built into the system

- Abstention rather than forced prediction. The system declines rather than guessing when its guarantee cannot be met.
- Distributional rejection. Inputs outside the training distribution are refused before classification.
- Uncertainty surfaced, not buried. Epistemic and aleatoric components are shown in the interface, and confidence is encoded visually as colour saturation so that certainty is legible without reading a number.
- Verified generation. The report writer cannot introduce a finding the vision model did not support.
- Audit trail. Every analysis, abstention, rejection, human review and rejected generation is recorded in an append-only log.
- Persistent non-diagnostic notice. Every surface of the application states that the system is not a medical device.

8.5 Intended and unintended use

The system is intended as a research prototype demonstrating uncertainty-aware triage. It is not validated for clinical use, has no regulatory clearance, and must not inform patient care. The most plausible misuse is deployment as a screening tool in a resource-constrained setting on the argument that an imperfect system is better than none. We reject that argument here: an unvalidated system with unknown subgroup performance can concentrate harm on precisely the populations such a deployment claims to serve.

9 Limitations

- Label noise. Labels were NLP-mined from free-text reports with roughly 10% error. No metric reported here can be more accurate than the labels it is measured against.
- Single-institution data. All data originates from one United States hospital system. Performance on other populations, equipment and acquisition practice is unvalidated.
- Marginal, not simultaneous, coverage. The conformal guarantee holds per label. Simultaneous coverage across all fourteen would require a multiplicity correction and is not claimed.
- Exchangeability. The guarantee assumes calibration and test data are exchangeable. Distribution shift — a new scanner, a new population — voids it, and the system has no mechanism to detect such shift beyond the OOD gate.
- Grad-CAM is correlational. Saliency indicates where activation correlates with a score, not why a decision was reached.
- Windowing is approximate. The viewer implements window and level as CSS filters over 8-bit images, which approximates but does not reproduce true DICOM windowing over 12-bit data.
- Uncertainty source varies. The live demonstration uses test-time augmentation because the published checkpoint lacks dropout. This is a robustness measure, not a Bayesian posterior, and is labelled as such in every response.
- Simulated triage environment. The reinforcement-learning results are obtained in simulation. No claim is made about behaviour in a real reading room.

10 Conclusion

We set out to build a chest radiograph triage system whose defining property is not accuracy but calibrated humility. The resulting system converts uncalibrated network scores into prediction sets with a distribution-free coverage guarantee, refuses inputs outside its training distribution, separates its own ignorance from genuine ambiguity in the image, and abstains rather than guessing when the guarantee cannot be met.

Building it required engaging with the full breadth of the unit: convolutional networks for classification, recurrent networks for longitudinal progression, adversarial networks for minority-class augmentation, variational autoencoders for distributional gating, transfer learning throughout, deep reinforcement learning for worklist ordering, transformers for architectural comparison, and generative language models for reporting — composed into one system rather than assembled as separate exercises.

The most valuable lessons were negative ones. An early triage environment could not distinguish a good policy from a random one and had to be reformulated. A pretrained checkpoint turned out to contain no dropout, silently invalidating the intended uncertainty method and forcing an honest alternative. The interface at one point displayed a probability above its threshold beside the label 'not included', because the stored scores and the scores used for the decision had diverged. Each was found by testing a claim the system made about itself, and each would have survived a demonstration that only checked whether the application looked correct.

That is the underlying argument of this project. A clinical system earns trust not by being confident but by being able to say, precisely and verifiably, what it does not know.

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Appendix A — Syllabus coverage map

Every topic in the unit outline is realised inside the single system rather than as a standalone exercise.

Wk	Unit topic	Realisation	Notebook
1–2	Neural networks, activations, backpropagation	Backprop from scratch in NumPy; gradient check; activation and optimiser ablation	01
3	Convolutional neural networks	DenseNet-121 multi-label classifier; conformal calibration	02
4	Recurrent neural networks	GRU over per-visit embeddings; persistence baseline	03
5	Long short-term memory	RNN/GRU/LSTM ablation; gradient-flow analysis	04
6	Generative adversarial networks	DCGAN minority-class augmentation; measured Δ AUPRC	05
7	Autoencoders and VAEs	Convolutional VAE OOD gate; latent traversal	06
8	Transfer learning	Scratch / frozen / full / progressive; data-efficiency curve	07
9	Deep reinforcement learning	Double DQN worklist triage vs FIFO, heuristic, oracle	08
10	Practical applications (ViT, CLIP)	ViT-B/16 vs CNN; BiomedCLIP zero-shot baseline	09
11	Generative AI integration	GAN + VAE + grounded LLM reporting with verifier	10
12	Ethical and societal impacts	Disaggregated audit; shortcut probe; model card	11

Table A1. Syllabus coverage.

Appendix B — Reproducibility

- Seed. All experiments use seed 20260812 for NumPy, PyTorch and CUDA.
- Splits. Patient-disjoint 70/10/20, asserted programmatically at load time.
- Environment. Notebooks run on Colab free tier (T4). No paid runtime is required.
- Weight provenance. The deployed demonstration uses TorchXRayVision pretrained weights so that the live system is functional from day one. This is stated in the README, the model card and the API health endpoint. No result produced by borrowed weights is presented as the group's own training outcome.
- Tests. 80 automated tests cover the conformal head, uncertainty decomposition, triage ordering, progression logic, grounding verification and the full API surface.
- Artefacts. Notebooks 02, 08 and 11 export `conformal_calibration.json`, `dqn_policy.json` and `fairness_report.json` respectively, which the deployed API loads. Until they are present, the API reports that its coverage guarantee is not in force.

— End of report —